

Code-Layout, Readability and Reusability

Date	6 july 2026
Team ID	XXXXXX
Project Name	Personalized network assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	yes	Proper indentation followed throughout the project.
2	Proper File Structure	Files and folders are logically organized	yes	separate folders for backend, frontend, models, services, and tests.
3	Meaningful Variable Names	Variables reflect their purpose clearly	yes	Clear variable names like event_description, conversation_history, themes.
4	Function / Method Names	Functions are descriptively named	yes	Functions are descriptive (e.g., extract_themes(), generate_response()).
5	Code Comments	Inline and block comments explain log c	partial	Comments added for important modules and functions.
6	Modular Design	Code is split into reusable functions/modules	yes	Backend, frontend, AI models, and APIs are organized into reusable modules.
7	No Redundant Code	Duplicate or unused code is removed	yes	Duplicate code removed using reusable functions.
8	Error Handling	Exceptions and errors are handled gracefully	yes	Exceptions are handled using try-except blocks and validation.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolog	Reused Where	Reusability Level (High / Medium / Low)
1	Authentication Module	Python, FastAPI	User Registration & Login	high

2	theme Extraction Module	DistilBERT, Python	Event Description Analysis	high
3	Conversation Generator	GPT-2, Python	Generates Networking Suggestions	high
4	Fact Checking Module	Wikipedia API	Response Verification	medium
5	Streamlit UI	Streamlit	User Interface & Dashboard	high

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure.
Readability	5	Clean, meaningful names and readable code.
Reusability	5	Modular components reused across the application
Documentation / Comments	4	Good documentation with scope for more inline comments.
Overall Score	4.8	Good quality, modular, and maintainable project.